

Microcontroller System Design Lab 0

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1 Introduction

The purpose of this preliminary lab is to begin to interface with the microcontroller and verify its functions using software. I have taken mechatronics, cse 100 and ece 13e and have prior experience with microcontrollers. For lab 0 I collaborated with Grant, Nadia and Tristen.

2 Hello World!

In the first part of the lab to verify the functions of the 8 LEDs and 4 push buttons of the PIC32MX340F512H microcontroller board we used MPLAB X IDE using the buttons as input and the LEDs as output. From previous experience with similar boards in classes such as CSE 13E we knew how to access the specific functions and bits of the board. We used the chipKIT Basic IO Shield Reference Design and the chipKIT™ Basic I/O Shield Board Reference Manual to provide the necessary information to confirm how to interface with the board.

2.1 LEDS

The LEDs are on Port E and the entire pattern can be reset with an 8bit value or individually set. The TRIS sets the bit to an input or output and the LAT is an output value that can be set manually. The LEDs toggle on and off for any button press.

```
#define LEDS LATE
#define LED_TRIS TRISE
```

Figure 1: LED defines

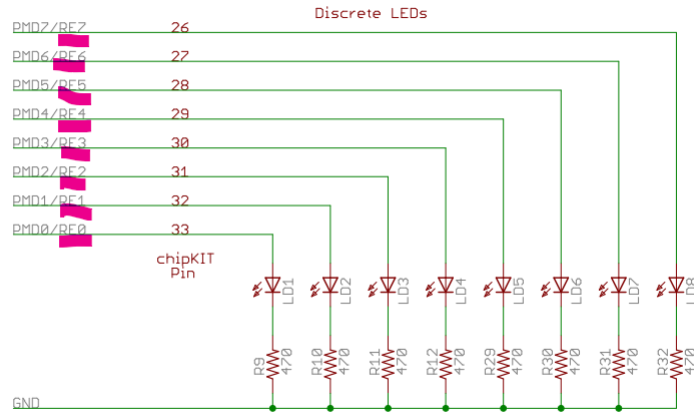


Figure 2: LED Schematic

2.2 Push Buttons

Consulting the schematic and header files the specific buttons were on slightly different ports and were accessed individually rather than as a simple single 8 bit value. For the functionality the buttons responded to an up event, being released from being pressed rather than just being pressed.

```
#define BTN1 PORTFbits.RF1
#define BTN2 PORTDbits.RD5
#define BTN3 PORTDbits.RD6
#define BTN4 PORTDbits.RD7
```

Figure 3: Push Button Defines

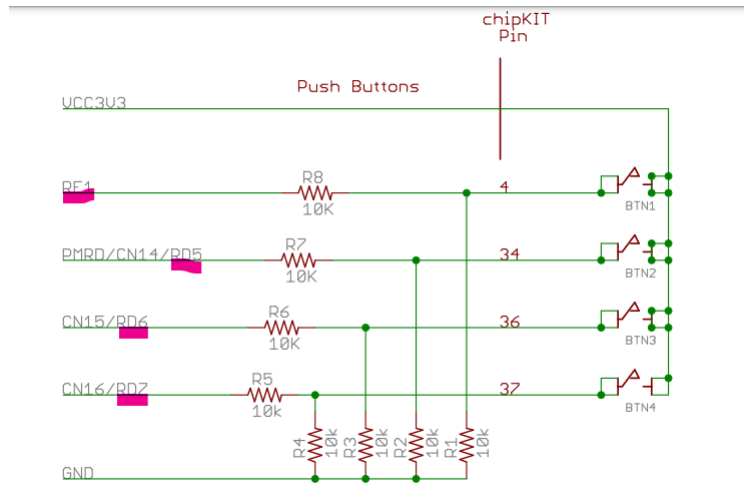


Figure 4: Push Button Schematic

3 Improved Hello World!

The second part of this preliminary lab was concerned with using timing to form a continuous while loop of binary addition and resetting itself for a complete loop and a button press.

3.1 Algorithm

I placed the pseudocode side by side in the IDE and translated it separating the parts of the lab with macro defines. The LEDs 8bit value incremented itself in binary continuously, using a delay function to time out each iteration of the while loop exactly. Initially I shifted the bit and then incremented it to make a sort of filling up bar of the LEDs but changed it back to simple binary addition to match the lab manual.

3.2 Timing

To get accurate timing measurements we used the delay function and the stopwatch feature of the IDE. We set the value of the nop loop arbitrarily and measured how many cycles it would take, settling on a value that was approximate to the necessary 5 ms. We attempted to ascertain a value from a mathematical method by taking into account the 80 MHz system of the clock, and decided to use a simpler and just as effective method of timing it out in real time to get an approximate value. To get the slower looping for the LEDs we added a value that would count the time and the LEDs would only be set when the time was exactly divisible by 250 using the modulo function through the iterations.

References

- [Manual] Diligent. (2011). the chipKIT™ Basic I/O Shield Board Reference Manual. Copyright Digilent, Inc.
- [Schematic] Diligent. (2011). chipKIT Basic IO Shield Reference Design. Copyright Digilent, Inc.